stated that including auction information in core data would eliminate proprietary data costs as a barrier to auction trading and encourage a broader range of market participants to submit trading interest into auctions, enhancing market liquidity and price discovery.[463] Another commenter cited recent market wide circuit breaker halts and the consequent re-opening auctions as reasons for its support of including auction information in core data.[464] Another commenter stated that retail investors should be properly informed with appropriate information about the indicative auction price and the trading imbalance.[465]

The Commission agrees with these comments on the growing significance of auctions and auction information to investors and other market participants. As the Commission stated in the Proposing Release, opening and closing auctions conducted by the exchanges have become increasingly important liquidity events and represent a significant proportion of overall trading volume.[466] The growth of passive, index-tracking investment strategies through mutual funds,

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II at 4; BestEx Research Letter at 5 (stating that competing consolidators should be able to determine whether or not to include auction information in the products offered to subscribers); Healthy Markets Letter at 3; Clearpool Letter at 15; Wellington Letter at 1; Proof Trading Letter at 1 (stating that auction information could be useful for agency trading); NovaSparks Letter at 1.

463 See ICI Letter at 9–10 ("Doing so also would eliminate proprietary data costs as a barrier to auction trading and encourage a broader range of market participants to submit trading interest."); see also SIFMA Letter at 7.

464 See RBC Letter at 5.

465 See Angel Letter at 8.

466 See Proposing Release, 85 FR at 16756 (stating that auctions account for approximately 7% of daily equity trading volume based on data available on Cboe's website from November 2019). Staff conducted the same analysis of auction data that was included in the Proposing Release but for a more recent time period (the month of June 2020) and observed that auctions account for approximately 7% of daily equity trading volume. Staff also observed that auctions accounted for more than 20% of total volume on two days (June 19 and June 26). See Cboe: U.S. Equities Market Volume Summary,

ETFs, and similar products has contributed to the higher concentration of trading in closing auctions.[467]

For these reasons, the Commission believes that auction information should be included in core data to promote more informed and effective participation in auctions by market participants and to potentially broaden the range of market participants who participate in auctions, enhancing auction liquidity and price discovery. Specifically, the Commission believes that auction information, such as order imbalances and indicative prices, helps market participants determine whether to participate in auctions, how to trade leading up to an auction, and how to best place their trading interest into an auction.[468] Finally, the Commission agrees that recent market wide circuit breaker halts, which occurred after the Commission's issuance of the Proposing Release, further underscore the need for auction information to be included in core data so that information related to the reopening auctions that occur after such halts is broadly disseminated to market participants, promoting more informed participation in these auctions.

(b) Asserted Violation of Intellectual Property Rights

One commenter stated that it "possesses copyright rights in its auction data as a compilation…" and the proposal would require that it "forfeit these copyrights rights."[469] The commenter noted that there have been "auctions of financial instruments for hundreds of years," and that it "has developed a unique approach to auctions that includes a creative selection and

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available at https://markets.cboe.com/us/equities/market_share/ (last accessed Aug. 30, 2020).

467 See Proposing Release, 85 FR at 16756–57.

468 See id. at 16826–27.

469 Nasdaq Letter IV at 51.

arrangement of auction data."[470] The commenter stated that it "has the exclusive right to reproduce its compilation in copies, to prepare derivative works based on the compilation, and to distribute copies of the compilation."[471] The commenter further stated that the proposal would "force Nasdaq to surrender these rights, robbing Nasdaq of its ability as copyright owner to obtain fair market value of licenses for its intellectual property."[472]

Some commenters disagreed.[473] One commenter stated that viewing auction information (or depth of book data) as the intellectual property of the exchanges would ignore the fact that the broker-dealers who submit the orders and the investors who generate the orders are the

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470 See id. ("When performing an opening or closing auction, Nasdaq receives orders and disseminates (via NOII messages) the results of those simulations. The frequency at which the NOII is disseminated changes over the course of an auction; for example, in the closing auction, the NOII is disseminated every ten seconds for the first five minutes of the auction, and then every second for the final five minutes of the auction. The NOII includes a number of data fields, including: Symbol (indicating the security to which the NOII relates); Near Indicative Price (which is based on orders in both the closing and continuous books); Far Indicative Price (which is based on orders solely in the closing book); Current Reference Price (which is based solely on orders in the continuous book); Paired Shares (indicating how many shares would execute at the Current Reference Price); Imbalance Shares (indicating the number of shares that would remain after execution at the Current Reference Price); and Imbalance Side (indicating whether the Imbalance Shares relate to buy orders or sell orders). The three prices in the NOII are not simply based on executed transactions, but rather they are simulations of what the price "would be" if the auction were to execute at that moment, based on different inputs. Each day, Nasdaq generates over 400 simulations of these three prices for each security. As there are over 3000 securities traded on Nasdaq each day, this means that Nasdaq compiles more than 1.2 million NOII records each day. The selection and arrangement of the NOII data fields are original and reflect Nasdaq's creative judgment; Nasdaq did not copy this selection of auction data, and there is no precedent for this unique and creative assembly of auction data fields….").

471 Id.

472 Id.

473 See Fidelity Letter at 5; SIFMA Letter at 7.

source of this data and would contravene broker-dealers and investors' ownership rights in the underlying data.[474]

The Commission does not agree that the definition of auction information or its inclusion in core data would violate any copyright interests of the commenter. The Commission has the authority to determine the content of the quotation and transaction information made available under the national market system rules, including information related to exchange auctions.[475] In addition, other Commission rules require the public disclosure or provision of information that must be compiled, such as Rules 601 (transaction reports), 602 (quotations), 605 (order execution information), and 606 (order routing information) of Regulation NMS and Exchange Act Rule 15c2-12 (credit rating and audit information related to municipal securities).[476]

Moreover, auction information, such as order imbalances and indicative pricing, pertains to certain outputs of an exchange's auction process. Such auction information does not require the disclosure of any details about the process of the auction or require the exchanges to "reproduce" their compilations. Rather, the exchanges can comply with the requirement to make information related to their auctions available to competing consolidators and self-aggregators without using their existing compilation systems; they are free to collect and publish the factual

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474 See SIFMA Letter at 7.

475 See supra note 114.

476 See 17 CFR 240.15c2-12. See also Securities Exchange Act Release No. 74244 (Feb. 11, 2015), 80 FR 14564, 14669–70 (Mar. 19, 2015) (Regulation SBSR—Reporting and Dissemination of Security-Based Swap Information Adopting Release) ("Under the federal securities laws, the Commission imposes a number of requirements that compel the provision of information to the Commission itself or to the public.… Businesses that operate in an industry with a history of regulation have no reasonable expectation that regulation will not be strengthened to achieve established legislative ends.").

information required by the amendments to be made public by any means they choose in a manner that does not utilize any copyrightable format or collection methodology. Therefore, to the extent that any intellectual property rights attach to exchange auction processes themselves, these amendments do not violate those rights.[477]

Furthermore, exchanges will continue to be compensated for making auction information available to competing consolidators and self-aggregators through fees established by the effective national market system plan(s). The Operating Committee of the effective national market system plan(s) will propose fees for the data content underlying consolidated market data, including auction information, as well as updates to the formula for allocating revenues from this data among the SROs.[478] In so doing, the Commission expects the Operating Committee to assess the impact of the inclusion of auction information in consolidated market data on the fees to be charged for the data content underlying consolidated market data, as well as how exchanges

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477 For example, the dissemination of auction information as required by these amendments does not violate any patents that the commenter has identified, as these amendments do not compel the disclosure of the process of how the auction prices are derived. See Rule 600(b)(5) of Regulation NMS (defining "auction information," by reference to information specified in exchange rules, which do not disclose how auction prices are derived). Rather, these amendments require exchanges to make auction information available to competing consolidators and self-aggregators so that market participants may submit trading interest into auctions in a more informed manner.

478 Auction data for a particular NMS stock will likely be generated by a single exchange, namely the primary listing exchange for that stock. However, all data elements that make up consolidated market data, such as individual quotes and trades or regulatory data, originate from a single SRO, and, as explained below, the Operating Committees of the effective national market system plans historically have determined how best to allocate consolidated market data revenues among the SROs to fairly reflect their individual contributions. See infra Sections III.E.2(b), III.E.2(f). In addition, certain NYSE auction data is currently included in Tape A. See Proposing Release, 85 FR at 16757.

that contribute auction information should be compensated for this data through the allocation formula.

(c) Assertion of Insufficient Justification for Inclusion of Auction Information in Core Data

One commenter characterized auction information as "esoteric" and "designed to assist sophisticated market participants," arguing that anyone who needs this information can buy it now, that it is likely to be "useless" to anyone who does not currently buy it, and that including it in core data will not make it "any more or less available."[479] Similarly, another commenter stated that auction information is not "necessary or helpful for all investors" and that its inclusion in core data would increase core data costs unnecessarily and create confusion as to what is necessary for regulatory compliance.[480]

A variety of commenters—including exchange, buy-side, and sell-side market participants—stated, and the Commission agrees, that auction information is not "esoteric" information that would be of use only to some small subset of "sophisticated" market participants.[481] Rather, the Commission continues to believe that auction information would be useful or beneficial to a broad cross-section of market participants—including retail investors, according to one commenter[482]—and would enable these market participants to participate in auctions, and to trade leading up to auctions, in a more informed manner.[483] Furthermore, as explained above, even market participants that do not directly acquire all elements of core data,

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479 See Nasdaq Letter IV at 33.

480 See TD Ameritrade Letter at 5.

481 See supra note 462.

482 See Angel Letter at 8.

483 See supra note 468.

including auction information, will still indirectly benefit from the inclusion of auction information in core data. The inclusion of auction information in core data will facilitate greater access to this information among a broader group of executing broker-dealers and will enable them to place orders into auctions more effectively and to achieve better executions for their customer orders.[484] As commenters stated, proprietary data costs may discourage some market participants from participating in auctions.[485] The Commission is sensitive to these concerns and believes that including auction information in core data would help facilitate its broader dissemination.[486] Finally, the Commission is not mandating the consumption of auction information and has explained the implications of these amendments on best execution obligations above.[487] The Commission believes the inclusion of auction information in core data will not create confusion as to regulatory requirements, as one commenter stated might happen.[488]

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484 See supra Section II.C.2(a); infra Sections V.C.1(c)(iii) (discussing how the amendments will affect access to auction information); III.E.2(c); V.C.2(b)(i)a (discussing how the amendments will affect data content fees).

485 See ICI Letter at 9–10 ("Including auction information in the consolidated feed would enhance transparency into market activity. Doing so also would eliminate proprietary data costs as a barrier to auction trading and encourage a broader range of market participants to submit trading interest."); SIFMA Letter at 7 ("Adding this [auction] data to the definition of core data would assist with alleviating some of the discrepancies in content between the exchange proprietary feeds and the current SIP feeds and provide market participants with the ability to rely on SIP feeds rather than incurring the substantial costs in being forced to purchase both the proprietary data and the SIP data.").

486 See supra note 484.

487 See supra Section I.E.

488 TD Ameritrade Letter at 4.

(d) Include Competing Crosses in Auction Information

One commenter recommended expanding the proposed definition of auction information to include data on competing crosses offered by national securities exchanges other than the listing market, such as one of the commenter's products (the Cboe Market Close product), so that investors have a full view of exchange trading in other mechanisms through which investors can seek to have their orders executed at official opening or closing prices.[489]

The Commission does not believe that information on competing crosses should be included in core data at this time. Auctions are held pursuant to exchange rules at specified periods during the trading day (e.g., at the open, at the close, or during the day to reopen a stock that has been halted) when continuous trading is not occurring. During auctions, buy and sell orders generally interact at the single price, within limits, that maximizes the trading volume that can be executed.[490] For example, a closing auction generally is held at the end of regular trading hours on the primary listing exchange pursuant to a process set forth in the primary listing exchange's rules to determine a security's official closing price.[491]

While the Cboe Market Close process seeks to provide executions on Cboe BZX Exchange at the official closing price published by the primary listing exchange, which is typically determined through an auction, it is not itself an auction process that establishes pricing.[492] In proposing to include auction information in core data, the Commission was

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489 See Cboe Letter at 21.

490 See, e.g., NYSE Rule 123C(8). See also Proposing Release, 85 FR at 16756.

491 See, e.g., NYSE Rule 123C(1)(e). See also Proposing Release, 85 FR at 16756.

492 Securities Exchange Act Release No. 88008 (Jan. 21, 2020), 85 FR 4726 (Jan. 27, 2020). Through Cboe Market Close, buy and sell market on close orders for stocks not listed on

responding to the growing importance of auctions themselves rather than competing cross processes that leverage auction-based pricing. Because competing cross processes are a derivative of the underlying auctions that establish prices, the Commission does not believe that including information regarding Cboe Market Close or similar competing cross processes would further the objective of promoting more informed participation in auctions. The Commission's decision does not preclude the dissemination of information related to such processes on a proprietary basis or prevent competing consolidators from acquiring and providing this information to their subscribers. Additionally, self-regulatory organization-specific program information can also be required to be included in consolidated market data pursuant to the national market system plan or plans required under Section 242.603(b) or amendments thereto that are approved by the Commission.[493]

Cboe are matched together and executed at the closing price of the stock's primary listing exchange. Because the Cboe Market Close process is using the primary listing exchange's closing price as the execution price, the Cboe process is not independently discovering a closing price different than the primary listing exchange. Id. at 4727, 4738 ("The Commission finds that . . . Cboe Market Close should not disrupt the price discovery process in the closing auctions of the primary listing exchanges. Importantly, Cboe Market Close will only accept, match, and execute unpriced MOC orders with other unpriced MOC orders (i.e., paired-off MOC orders). Contrary to some commenters' assertions that MOC orders contribute to the determination of the official closing price, the Commission believes that paired-off MOC orders, which do not specify a price but instead seek to be executed at whatever closing price is established via the primary listing exchange's closing auction, do not directly contribute to setting the official closing price of securities on the primary listing exchanges but, rather, are inherently the recipients of price formation information.").

493 See supra Section II.B.2; infra Section II.K.2.

(e) Classification of Auction Information

One commenter stated that auction information (along with odd-lots and depth of book data) should be part of the consolidated market data, rather than core data, and expressed its doubt that firms would have an option to receive only a portion of core data.[494] Another commenter stated that auction information should be further split into three products: (1) core data that has no auction data but includes Cboe Market Close orders in order not to drag the processing speed of normal data distribution; (2) separate subscription for auction imbalance information (matched quantity, imbalance size, near price, far price, paired shares, and imbalance shares); and (3) integrated auction data in a combined feed, the speed of which may be slower than (1) and (2).[495]

The Commission believes that auction information should be part of core data. Core data includes elements that the Commission has determined to be useful to inform trading decisions by today's investors. Auction information is important to investors who wish to participate in the opening, reopening, and closing auctions, which make up an increasing proportion of overall trading volume. However, as discussed above,[496] the Commission is not creating any new regulatory obligation to consume auction information.[497] Furthermore, the Operating Committee

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494 See TD Ameritrade Letter at 6 ("Inclusion under the broader Consolidated Market Data definition would still require for the collection, aggregation, and dissemination of the data to make it available to self-aggregators and competing consolidators but would avoid future confusion about what is required of end users for regulatory purposes in the future.").

495 See Data Boiler Letter I at 20.

496 See supra Section II.C.2(a).

497 See supra Section I.E. These amendments do not mandate the consumption of auction information, regardless of whether auction information is defined as core data or consolidated market data, contrary to what one commenter suggested. See supra note 494 and accompanying text.